Investment Demand Shock and the Business Cycle

New Results

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What is new in this version

Five new results that dig into the baseline mechanism and its fragility:

1. Where does the high **Labor input** variance come from?
2. Are the **inflation and wage** signs (IST vs. ID) stable across parameters?
3. Why is the estimated **consumption** impact negative, while theory predicts positive?
4. How does a full **Smets–Wouters (2007)** model change the picture?
5. How does an **i.i.d. (independent)** investment demand shock change the result?

Notation: ε_s = IST shock, ε_d = investment demand (ID) shock, ε_m = MEI shock, ε_a = technology, ε_R = monetary, ε_g = government, $\varepsilon_b, \varepsilon_p, \varepsilon_w$ = preference / price- / wage-markup.

1. Why model hours are too volatile

- Baseline matches most moments, but $\mathrm{Var}(\hat{N}_t)$ is far above the data (see model-fit table).
- Decompose the log-linear labor demand/supply block:

$$\hat{N}_t = \frac{\hat{mc}_t + \alpha \hat{k}_t^s + \hat{\mu}_{a,t} - \theta_w \sigma \hat{c}_t}{\alpha + \theta_w \sigma_\ell}.$$

- R^2 of each component on $\hat{N}_t$: **consumption co-movement explains** $\approx 82\%$ of hours volatility; all other channels are negligible.

Component	$\mathrm{corr}(N_t, x_t)$	R^2 (%)
Consumption, $\hat{c}_t$	+0.907	82.29
Marginal cost, $\hat{mc}_t$	+0.061	0.37
Capital services, $\hat{k}_t^s$	-0.030	0.09
Technology, $\hat{\mu}_{a,t}$	+0.087	0.76

1. The root cause is volatile consumption

- Model $\text{Var}(\hat{c}_t)$ (level) is 39.2 vs. 2.8 in the data, and this is not because consumption is excluded from the baseline data set.
- The **same pathology exists in Smets–Wouters (2007)**: their level variance of consumption is 33.8 (computed from their replication package).
- Difference is hours: SW (2007) fits $\text{Var}(\hat{N}_t)$ much better because the **wage markup shock** $\mu_{w,t}$ loosens the wage–MRS link:

$$\hat{\pi}_t^w = \beta \mathbb{E}_t \hat{\pi}_{t+1}^w - \kappa_w \underbrace{\left(\hat{w}_t - \hat{mrs}_t + \frac{1}{\kappa_w} \mu_{w,t} \right)}_{\text{wage markup}}.$$

- Without that shock, the wage is nearly static, so consumption volatility passes straight into hours.

Variable	Smets and Wouters [2007]	Baseline model	Data
Consumption ($\hat{c}_t$)	33.75	39.23	2.80
ρ_1	0.993	0.996	0.901
Hours worked ($\hat{N}_t$)	8.85	32.22	6.29
ρ_1	0.975	0.993	0.907

2. Is the inflation/wage sign stable across parameters?

Question: the IRFs show *deflationary IST* and *inflationary ID*. How robust are these signs to the parameters?

- Inflation = PDV of marginal cost: $\hat{\pi}_t = \kappa_p \sum_{j \geq 0} \beta^j \mathbb{E}_t[\widehat{mc}_{t+j}]$.
- Two marginal-cost channels: wage and capital rental rate.

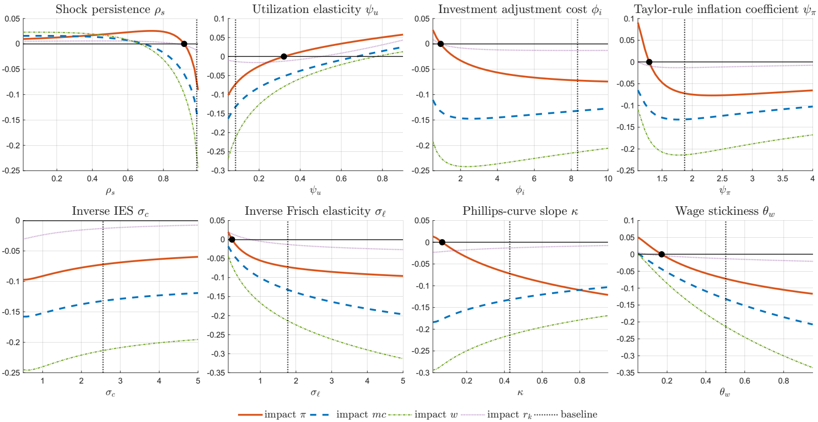
IST shock ε_s

- Inflation sign **not robust**: flips to inflationary in many cases.
- But the **wage sign is stable** (negative) across the sweep.

ID shock ε_d

- Inflation **robustly positive** across all 500 posterior draws and every parameter sweep.
- Wage **robustly positive** as well.

2. IST: inflation sign flips, wage sign does not



Each panel varies one parameter over a wide grid. The impact *inflation* sign is fragile; the *wage* sign stays negative.

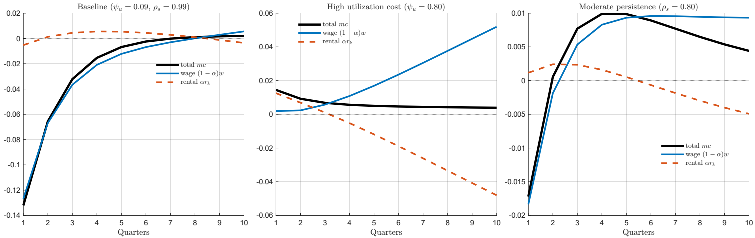
2. The role of ψ_u: why the IST inflation sign flips

ψ_u is the curvature of the utilization-cost function. The utilization FOC plus the firm's cost minimization give

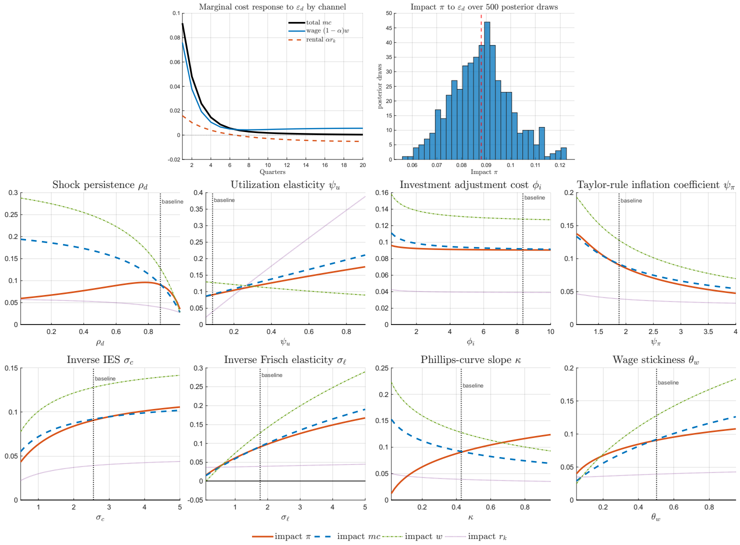
$$\hat{r}^k_t = \frac{\psi_u}{1-\psi_u}\,\hat{u}_t, \qquad \hat{r}^k_t = \psi_u(\hat{w}_t + \hat{N}_t - \hat{k}_{t-1}),$$

so $\hat{r}^k_t$ moves only a **fraction** ψ_u of $(\hat{w}_t + \hat{N}_t)$.

- **Low ψ_u (baseline ≈ 0.1):** rental channel muted; inflation is driven by the wage, which falls under IST $\Rightarrow$ **deflation**.
- **High ψ_u:** larger $a''(1)$ $\Rightarrow$ steeper adjustment cost $\Rightarrow$ capital supply can't adjust; rising labor pushes $\hat{r}^k_t$ up so the rental channel dominates $\Rightarrow$ **inflation**.

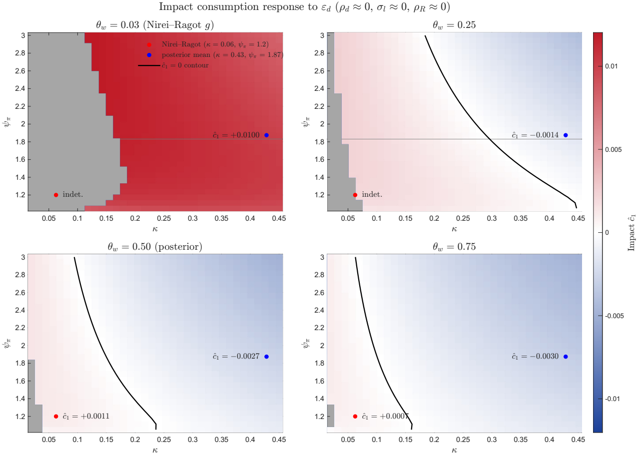


2. ID: inflation and wage are robustly positive



3. Why estimated consumption response is negative

- In the estimated baseline, the impact consumption response is negative, and **no single** parameter change reverses it. A positive response needs a **combination**.
- Aligning with Nieri and Ragot's calibration and varying κ_p, ψ_π, θ_w jointly: consumption turns positive only when **all three are low**.



3. The mechanism: kill the short-run substitution effect

Intuition: in the long run an ID shock just accumulates capital $\Rightarrow$ households are wealthier $\Rightarrow$ they consume more today. A positive impact response requires *shutting down* the short-run intertemporal substitution channel.

- Low $\theta_w \Rightarrow$ wage barely moves $\Rightarrow$ marginal cost barely moves.
- Low κ_p (flat Phillips curve) $\Rightarrow$ inflation barely moves.
- Low ψ_π (inactive Taylor rule) $\Rightarrow$ nominal rate barely moves.

Stable real interest rate $\implies$ no shift of c_t to the future $\implies$ **positive** $\hat{c}_1$.

4. Smets–Wouters (2007) model: what changes

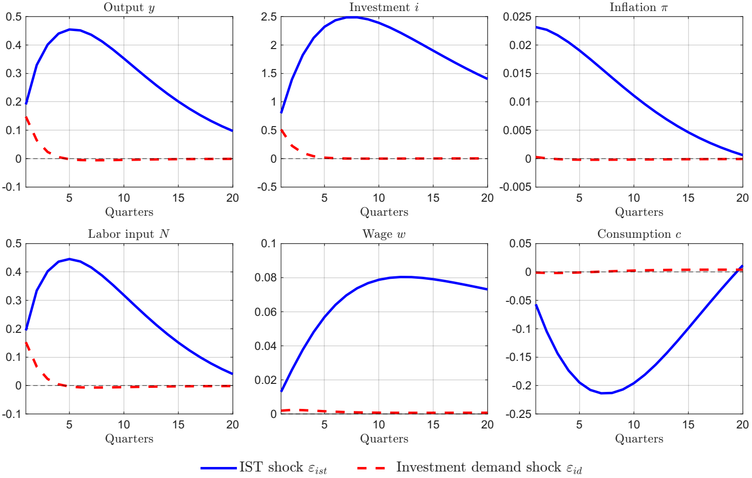
Expand the baseline with: habit formation, monopolistic labor market (Erceg-type wage Phillips curve), and preference / price-markup / wage-markup shocks; add consumption and wage growth as observables.

- **Better hours fit:** $\mathrm{Var}(\hat{N}_t)$ drops from 32.2 to 9.8 (data 6.3) — the wage markup shock does the work.
- **ID shock becomes marginal:** only 4.4% of output and 13.6% of investment FEVD; near-zero for everything else.
- Estimated $\kappa_p \approx 0.03$, $\kappa_w \approx 0.02$ (very flat) — partly propped up by the markup shocks.

Unconditional FEVD (%): Smets–Wouters-style model.

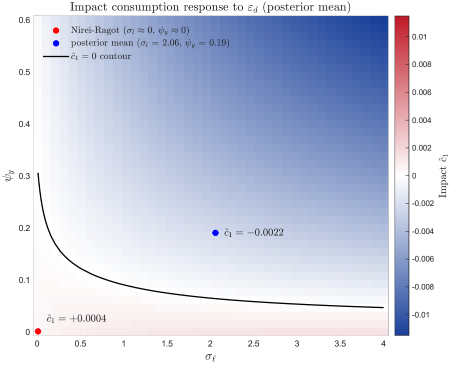
Observable	ε_g	ε_s	ε_R	ε_a	ε_b	ε_p	ε_w	ε_d
Output	6.15	10.31	7.86	43.58	17.21	5.00	5.52	4.37
Inflation	0.05	1.78	0.82	18.85	0.40	53.05	25.04	0.00
Interest rate	0.19	7.16	30.37	26.67	4.51	10.39	20.72	0.00
Hours	2.00	20.43	7.10	34.85	7.86	1.10	26.35	0.30
Investment	0.25	54.98	2.43	18.59	3.19	0.62	6.33	13.62
Wage	0.00	0.12	0.14	18.25	0.30	16.96	64.23	0.00
Consumption	0.25	2.36	6.48	43.49	42.01	1.92	3.49	0.00
Gov. spending	99.90	0.01	0.01	0.05	0.02	0.01	0.01	0.00

4. SW model: IRFs flip relative to the baseline



- **(i)** ID shock now generates *much smaller* fluctuations than IST (smaller estimated s.d.).
- **(ii)** IST shock now *inflationary*, with a *positive* wage response (sign flip vs. baseline).
- **(iii)** Consumption response to ID is much smoother: $\hat{c}_1$ is only 0.2% of $\hat{\imath}_1$ (vs. 12% in baseline).

4. SW model: consumption sign revisited



Now sweep σ_ℓ (inverse Frisch) and ψ_y (output-gap coefficient), because the flat Phillips/wage curves already sit near the Nieri and Ragot calibration.

Same logic: a positive $\hat{c}_1$ needs a low real rate.

- Small ψ_y: directly lowers Taylor-rule output-gap feedback.
- Small σ_ℓ: suppresses the real wage $\Rightarrow$ smaller wage gap $\Rightarrow$ smaller output gap.

5. Independent (i.i.d.) investment demand shock

Households learn the realized investment demand next period and undo it, so the shock may be *transitory*. Set $\rho_d = 0$, keep everything else at baseline.

- ID shock contributes 22.95% **of output** and 15.18% **of investment** FEVD.
- Down from $\approx 40\%$ / $\approx 50\%$ in the baseline: zero persistence cuts off the future influence.

Unconditional FEVD (%): independent (i.i.d.) investment-demand model.

Observable	ε_g	ε_s	ε_R	ε_a	ε_d
Output	6.67	38.60	9.35	22.42	22.95
Inflation	0.40	52.00	12.87	34.37	0.37
Interest rate	0.37	74.14	3.74	21.66	0.09
Hours	2.84	92.22	0.12	4.53	0.29
Investment	0.25	73.06	0.20	11.31	15.18
Gov. spending	99.59	0.17	0.04	0.10	0.10

Summary of new results

1. **Hours:** excess volatility comes almost entirely ($R^2 \approx 82\%$) from over-volatile consumption; a wage markup shock cures the hours fit.
2. **Signs:** ID inflation/wage are *robustly positive*; IST inflation sign is *fragile*, though its wage sign is relative stable.
3. **Consumption:** estimated impact is negative; a positive impact needs jointly low $\kappa_p, \psi_\pi, \theta_w$ (a stable real rate).
4. **SW (2007):** fits hours, but shrinks the ID shock to a marginal role and flips IST to inflationary.
5. **i.i.d. ID:** importance falls (output $\approx 23\%$, investment $\approx 15\%$) but the shock remains a meaningful output driver.

Takeaways

(i) SW model shows stronger ability to fit with the data, and delivers posteriors indicating a stickier economy.

(ii) Positive consumption impact require strict parameter values, which is not easy to reproduce in the estimation, but the logic of reducing short-run intertemporal substitution effect works.

References I

M. Nieri and X. Ragot. The Origins and Propagation of Animal Spirits Shocks. *working paper*, 2025.

F. Smets and R. Wouters. Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach. *American Economic Review*, 97(3): 586–606, 2007. ISSN 0002-8282.